Category Semantic-Guided Unsupervised Domain Adaptation Network for Hyperspectral Image Classification

Binge Cui, Shihan Huang, Guangbo Ren, Haojie Li, and Jianzhi Yu

***Abstract*— Domain adaptation methods enable model migration and adaptation across different domain data distributions. However, the source and target domains of hyperspectral images (HSIs) have large spectral offsets and spatial distribution differences, making the extraction of high-quality domain-invariant features between different domains essential for classification. To achieve a more consistent feature representation for each category between the source and target domains, we propose a category semantic-guided network (CSGNet) unsupervised domain adaptation (UDA) for HSI classification. CSGNet is designed to learn cross-domain invariant representation from category semantic information. First, to embed category semantic prior knowledge during feature learning, we extracted textual semantic features from the textual descriptions for each category and projected visual features into the semantic space via visual-linguistic alignment. A category representation memory pool is then introduced to store the visual-linguistic representations of different categories. Second, we propose a biclassifier adversarial learning (BCAL) method designed to generate inconsistent category predictions in the unlabeled target domain, thereby enhancing the classifier's discriminative capability regarding those hard-to-transfer features. Finally, to utilize the domain-invariant features stored in the category memory pool, a category attention (CA) module is proposed to guide the model's adaptation to the data from different domains, mitigating the impact of the differences in the domain data distributions. Extensive experimental results validated on three cross-domain datasets demonstrate that the proposed method outperforms other state-of-the-art methods. The source code is available at http://github.com/cuibinge/CSGNet**

***Index Terms*— Adversarial learning, category attention (CA), category-label semantic guidance, domain adaptation, domain-invariant features, hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL images (HSIs) consisting of hundreds of contiguous spectral bands are capable of capturing subtle surface information (e.g., changes in land cover type, growth of vegetation, and water content of soil). The HSI classification task aims to analyze the rich spectral–spatial information of HSIs to accurately identify the various categories of land cover on the surface. Therefore, HSI classification tasks have been widely used in urban planning [1], environmental monitoring [2], resource management [3], and other Earth observation applications.

With the advancement of machine learning, researchers have begun to introduce machine learning algorithms into the field of HSI classification, and common classifiers include K-nearest neighbor (KNN) classifier [4], support vector machine (SVM) [5], maximum likelihood classification (MLC) [6], random forest (RF) [7], and multinomial logistic regression (MLR) [8]. However, these methods can extract only low-dimensional spectral–spatial feature information and require manual feature design; thus, they may be limited in processing complex HSIs. In recent years, the development of deep learning techniques has led to new breakthroughs in HSI classification. Deep learning methods effectively improve classification performance by automatically learning feature representations through deep neural networks [9]. Among them, a stack autoencoder combined with logistic regression classifier was first applied to HSI classification [10], and better results were obtained. Since then, 1-D convolutional neural networks [11] and recurrent neural networks [12] have been used for spectral feature classification, further enhancing classification accuracy. Additionally, 2-D [13] and 3-D convolutional neural networks [14], as well as graph feature-enhanced networks [15], [16], were utilized to extract spatial features to reduce classification uncertainty. Wang et al. [17] proposed Flex-MCFNet, a multistage information complementary fusion network based on flexible mixup. The information between complementary modalities is fully interacted and enhanced through multiple stages of information complement and fusion. Wang et al. [18] introduced a novel multiscale interactive fusion network (MIFNet) for the joint classification of hyperspectral and SAR images, enhancing classification accuracy and feature fusion through innovative MIIE and GDFM modules. Duan et al. [19] proposed a novel multiple neighborhood-aware nonlinear collaborative analysis (MNNCA) method that designs a dual-level neighborhood reconstruction to search for optimal neighbors and mine the common attributes within the neighborhood and explores the collaborative relationship among multiple neighborhood sets

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in the nonlinear space of HSI data. Zhang et al. [20] developed a spatial-logical aggregation network (SLA-NET) for tree species classification in remote sensing images, enhancing precision and efficiency by fusing spectral, spatial, and morphological features with multiscale morphological feature extraction and adaptive feature fusion techniques.

All of the above methods involve single-scene classification methods for HSIs. However, in practical scenarios, the acquisition of HSIs from different years or different geographic locations is inevitably affected by a variety of factors, such as weather and light, which leads to differences in the spectral reflectance between the source-domain training samples and the target-domain test samples for the same land cover category [21]. Therefore, a single-scene classification model trained on one scene may not achieve satisfactory classification results on another scene due to spectral offsets. Transfer learning [22] focuses on how to apply experience and data features from a known domain to a completely new domain when there are insufficient labeled samples. The unsupervised domain adaptation (UDA) technique involves a subtype of transfer learning [23] that acquires HSIs from two regions, one called the source scene and the other called the target scene. Training using images of the source scene with labels and images of the target scene migrates the knowledge of the labeled source scene, thus improving the classification performance of the unlabeled target scene [24]. Domain adaptation methods make it possible to reuse the labeling information of similar samples across scenes, which is of great significance for HSIs to obtain efficient classification results in a realistic situation where labeled samples are scarce and challenging to obtain [25].

In recent years, several UDA methods have made significant progress in HSI classification. Ganin et al. [26] employed a domain adversarial neural network (DANN) through adversarial learning, which is effective in reducing the distributional differences between source and target domains. Nevertheless, for some specific target-domain data distributions, deterioration of feature recognizability may occur. Zhang et al. [27] proposed LDGnet, a novel HSI classification framework that integrates visual and linguistic modalities, leveraging prior knowledge in language to enhance cross-scene generalization through a dual-stream architecture and supervised contrastive learning. Zhang et al. [28] introduced TSTnet, a novel approach in HSI classification that integrates topological structure and semantic information through graph convolutional networks and consistency constraints, significantly enhancing cross-scene classification performance. Huang et al. [29] proposed the TAADA network, a novel approach in remote sensing image interpretation that leverages a dual-branch feature extraction strategy coupled with adversarial domain adaptation to enhance the classification of HSIs across different domains. Zhang et al. [30] proposed a graph information aggregation cross-domain few-shot learning framework, which effectively captures nonlocal relationships within classes and similarities across domains by combining graph information aggregation and domain adaptation strategies. Zhang et al. [31] proposed a single-source-domain expansion network that utilizes generative

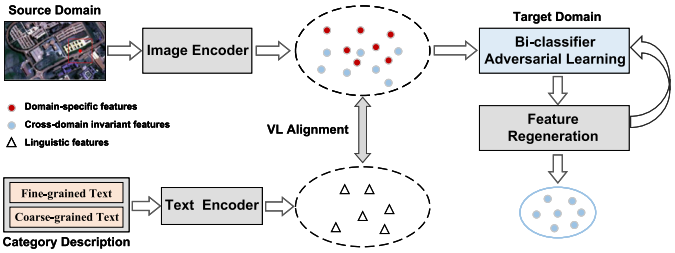


Fig. 1. Motivation of our CSGNet UDA.

adversarial learning and supervised contrast learning to ensure the reliability and validity of the extended domain, demonstrating superior performance in cross-scene HSI classification and multispectral image processing. Zhao et al. [32] proposed IDPNet, an intermediate domain prototype class-level learning network that utilizes dynamically generated intermediate domain features to construct class prototypes while incorporating interclass information into the prototype construction, achieving the class-centered distribution alignment for adaptation. Zhang et al. [33] proposed discriminative cooperative alignment of subspace and distribution to cooperatively reduce the geometric and statistical shift. Although these advanced adversarial methods simultaneously consider domain alignment and classification tasks, their motivation is to provide additional inductive information for the discriminator at the domain or decision level. Aligning task-irrelevant features may compromise the transfer learning results, leading to a negative transfer of the model and reduced classification performance.

This article proposes a category-label semantic-guided UDA framework for cross-scene HSI classification to address the above issues, as shown in Fig. 1. UDA methods involve spectral offset and spatial distribution differences between the source and target domains, making it difficult for conventional methods to fit the characteristics of the target domain. By incorporating textual descriptions of category labels, the feature learning process is guided to capture a consistent representation of cross-domain data. Category labels provide semantic information about feature categories, helping the model to understand and discriminate between different categories of features, thus allowing it to better adapt to the features of the target domain. UDA methods have difficulty effectively performing knowledge migration, preventing the model from accurately categorizing the target domain. With the category memory, the inherent invariant feature representation of each category can be stored, and the inherent representation of the category features can be effectively learned through an adaptive memory update mechanism. In this way, the model can migrate knowledge from the category features in the source domain to the target domain, thus improving the classification performance in the target domain. The main contributions of this article are as follows.

1) This article proposes CSGNet, an unsupervised model for HSI classification domain adaptation, which combines visual-linguistic alignment and biclassifier adversarial learning (BCAL) to enhance cross-domain classification accuracy and robustness. The alignment reduces distribution discrepancies (e.g., spectral offset

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and spatial differences) and helps the model better understand category meanings. BCAL further improves feature discrimination and generalization, significantly boosting performance in complex cross-domain HSI tasks.

2) The study introduces a category attention (CA) module with a category memory pool that stores domain-invariant feature representations for each category. This memory pool ensures consistent features across domains through dynamic updates, and the attention module helps focus on inherent category features, improving feature alignment and adaptation to distribution differences.
3) CSGNet employs an alternating two-phase adversarial training framework that decouples cross-modal alignment and domain-invariant feature learning. This framework alternates between source-domain semantic alignment and target-domain adversarial adaptation, ensuring stable generalization and avoiding modality conflicts in end-to-end methods.

The remainder of this article is organized as follows. Related work regarding domain adaptation and image–text models is introduced in Section II. Section III provides a detailed introduction to the proposed CSGNet. Extensive comparative experiments and ablation studies are conducted in Section IV, and the conclusion is presented in Section V.

II. Related Work

A. Domain Adaptation

Domain adaptation methods are essential tools for addressing spectral offset in HSI classification. Researchers aim to find ways to make the feature representations of the source and target domains more similar, enabling better performance in the target domain without using label information from the target domain. Domain adaptation methods achieve domain-invariant feature representations through either feature matching or adversarial learning.

Feature matching methods aim to reduce the differences in feature distributions by minimizing certain domain dissimilarity metrics. The maximum mean discrepancy (MMD) method recalibrates the domains by measuring the distribution divergence between the two domains to mitigate spectral offset [34]. The heterogeneous transfer convolutional neural network (HTCNN) leverages a pretrained CNN model to extract generic and discriminative features from the ImageNet dataset. It addresses channel mismatch issues through mapping layers and attention mechanisms, using these features to initialize the HSI classification network [35]. The deep subdomain adaptation network (DSAN) focuses on precisely aligning the distributions of relevant subdomains within different domains. It employs a local MMD (LMMD) loss, eliminating the need for adversarial training and enabling rapid convergence [36]. The confident learning-based domain adaptation (CLDA) method combines confident learning and domain adaptation in a framework that enhances the discriminative capability of a neural network by selecting high-confidence target samples from pseudolabels [37]. The domain adaptation with discriminative distribution manifold embedding (DA-DDME) method employs neural networks to learn the embedding space of images. The authors minimize distribution differences to adjust the classifier while preserving the original structure of the embedding space to prevent distorted mappings in the target domain to ensure the classifier's effectiveness in both domain [38]. However, in the feature matching process, discriminative feature extraction heavily relies on source-domain data [33], and the contribution of target-domain data to discriminative representations is minimal.

Adversarial learning methods employ adversarial learning strategies to train models to produce domain-invariant features. The dynamic adversarial adaptation network (DAAN) dynamically assesses the relative importance of global (marginal) and local (conditional) domain distributions to improve domain adaptation more effectively [39]. The deep metric learning-based feature embedding (ED-DMM-UDA) method introduces a feature embedding model based on deep metric learning and adds it to an adversarial framework to adapt feature distributions. It can learn similarity under a few labeled samples and achieve knowledge transfer from the source to the target scene through UDA [40]. The deep cross-domain few-shot learning (DCFSL) method addresses domain offset problems via conditional adversarial domain adaptation. It performs few-shot learning in both source and target classes, enabling the discovery of transferable knowledge in source classes and the learning of discriminative embedding models for target classes [33]. The maximum classifier discrepancy method attempts to align distributions of source and target by utilizing the task-specific decision boundaries [41]. Unlike previous adversarial learning methods that use domain discriminators, biclassifier domain adaptation methods use two different classifiers instead of a domain discriminator [42], [43].

B. Image–Text Foundation Models

The image–text models perform pretraining by learning a strong joint representation of text and images from a large-scale collection of image–text pairs, exhibiting remarkable transferability across various downstream vision and language tasks. Image–text foundation models can be categorized into two primary architectures based on their design: early fusion single-stream models and late fusion dual-stream models [44].

Early fusion single-stream models fuse two inputs of combined data from different modalities to generate a new data type. For example, VisualBERT [45] consists of a series of transformer layers that implicitly align elements of input text with relevant image regions using self-attention mechanisms. UNIMO [46] aligns text and image information into a unified form by utilizing a large-scale text corpus and image data and employing cross-modal comparative learning (CMCL) to semantic space, modeling image and text representations in a single converter of a single semantic space.

In contrast, the late fusion dual-stream models employ separate image encoders and text encoders for advanced feature representation before integration. Wang et al. [47]

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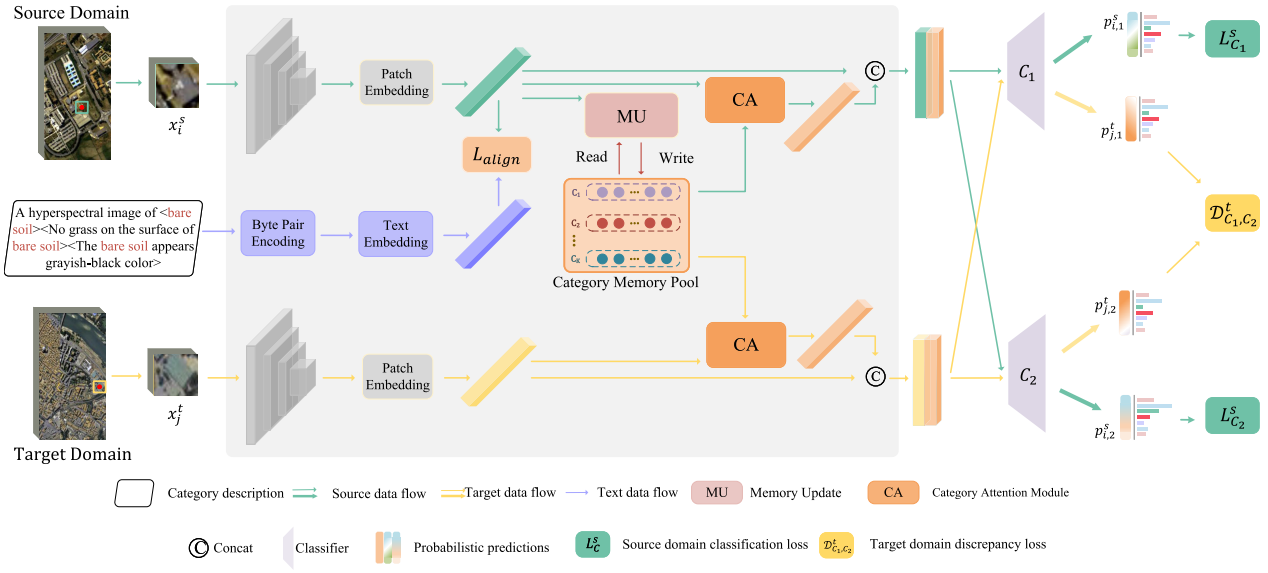


Fig. 2. Overall framework of the proposed CSGNet.

proposed a domain generalization method, EHSnet, which enables zero-shot generalization for cross-scene HSI classification. By leveraging multilayered explicit high-level semantic (EHS) information extracted from various textual sources, the EHS network (EHSnet) provides precisely relevant semantic guidance to the image encoder. Through the alignment of text and image features within a shared semantic space, this approach enhances the image encoder's representation capability, improving its ability to adapt across different domains. CLIP [48] achieves zero-sample migration by pretraining a model to understand associations between images and related text. Coca [49] seamlessly integrates natural language supervision by pretraining an end-to-end approach to training from scratch on large-scale web text data and images for representation learning. CLIP, DeCLIP, and Coca employ contrastive learning as their training objective. The core principle of contrastive learning is to automatically construct similar pairs of positive samples and dissimilar pairs of negative samples to bring the positive pairs closer together and the negative pairs farther away in the projection space.

III. Proposed Method

The proposed category semantic-guided network (CSGNet) for unsupervised domain adaptive HSI classification, as shown in Fig. 2, consists of three main parts: learning domain-invariant features for land cover categories, storing and utilizing land cover category representations, and adversarial learning between biclassifier and the encoder. First, by embedding the textual descriptions of each land cover category, we captured the deep-level semantic representations of these categories. Second, we conducted embedding learning on the source-domain hyperspectral pixels and semantically aligned the hyperspectral pixel embeddings with the text embeddings to obtain the domain-invariant features for each category, which will be introduced in Section III-A. These domain-invariant features of land cover categories are used to update the category representations in the category memory pool during each iteration, with the update process introduced in Section III-B. Subsequently, the CA module was employed to compute the relevance between the embedding of each target-domain hyperspectral pixel and the representations of each category in the category memory pool. It then dynamically fused the relevant category representations to supplement the feature representation of the target-domain hyperspectral pixel, which will be detailed in Section III-C. Finally, drawing inspiration from the adversarial game between the generator and discriminator in generative adversarial networks, we introduced two classifiers. They were tasked with making inconsistent category predictions in the unlabeled target domain, aiming to identify whether the features generated by the encoder contain hard-to-transfer source-domain-related features, as elaborated in Section III-D. During the inference stage of the model, we utilized the joint prediction of the two classifiers in the target domain as the final inference result.

A. Learning Domain-Invariant Features for Land Cover Categories

Traditional category labels are usually represented using one-hot vectors. However, these vectors lack the deep-level semantic information associated with the categories and can only offer limited supervision for extracting the discriminative features of the samples. In contrast, the textual descriptions of category labels are independent of the intrinsic features of the image domain and contain rich semantic information. Therefore, they provide powerful guidance for domain adaptation.

Each land cover category in the HSI corresponds to a coarse-grained textual description and two fine-grained textual descriptions. The coarse-grained text is formulated in a fill-in-the-blank format, using "A hyperspectral patch of <category name>" as a standard for each category. The fine-grained text is derived from amalgamating preexisting knowledge and expert-provided information (e.g., color, shape, and spatial

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relationships) regarding the land cover elements, such as "The surface of the meadow is green with grass." The byte-pair encoding and textual embedding modules transform these textual descriptions into a processable vector form. Concurrently, the image encoder extracts both the spectral and contextual features of the hyperspectral pixels, while the patch embedding module converts these spatial–spectral features into embedding vectors with the same dimensionality as the category semantic features. Subsequently, through correlation modeling, we can align the spatial–spectral features of the hyperspectral pixels with the category semantic features in order to learn the domain-invariant features for the land cover categories.

In CSGNet, the image encoder and text encoder are jointly trained. The alignment loss of image to coarse-grained text, denoted as L_{coarse}, represents the loss function that measures the alignment between the image features and the coarse-grained text representations, as depicted in the following equation:

$$L_{\text{coarse}} = -\sum_{i=0}^{N_s} \frac{1}{|P(i)|} \left(\sum_{p \in P_l(i)} \log \frac{\exp\left(\mathbf{v}_i^T \mathbf{l}_p^+ / \tau\right)}{\sum_{n \in N_l(i)} \exp\left(\mathbf{v}_i^T \mathbf{l}_n^- / \tau\right)} + \sum_{p \in P_v(i)} \log \frac{\exp\left(\mathbf{l}_i^T \mathbf{v}_p^+ / \tau\right)}{\sum_{n \in N_v(i)} \exp\left(\mathbf{l}_i^T \mathbf{v}_n^- / \tau\right)} \right) \tag{1}$$

where each visual feature v_i and text feature l_i represents the embedding of sample i in the minibatch. $P_v(i)$ and $N_v(i)$ are the positive and negative sample sets of the visual feature, respectively, while $P_l(i)$ and $N_l(i)$ are the positive and negative sample sets of the text feature, respectively. $|P(i)|$ signifies the number of positive samples, with $|P_v(i)| = |P_l(i)| = |P(i)|$. v_p^+ and v_n^- represent a positive and negative visual sample, respectively, while l_p^+ and l_n^- represent a positive and negative text sample, respectively. N_s denotes the size of the source-domain dataset, while the temperature parameter τ regulates the smoothness of the similarity. Similarly, the alignment loss of image to fine-grained text, denoted as L_{fine}, is structured to be similar to L_{coarse}.

The alignment loss of image to prior knowledge containing coarse-grained text and fine-grained text, denoted as L_{align}, as depicted in the following equation:

$$L_{\text{align}} = (1-\alpha) L_{\text{coarse}} + \alpha L_{\text{fine}} \tag{2}$$

where α represents the contribution of the fine-grained text feature.

B. Category Representation Memory Update

To store the representations of the different categories, we introduced a category representation memory pool. The pool size is $L \times K$, where L denotes the length of each category representation and K is the number of categories. During the initial training stage, the category representations in the memory pool are initialized as the average of the deep-level semantic features of the hyperspectral pixels for the respective categories.

For each iteration, we dynamically updated the category representations using the aligned land cover category representations. Let R_{C_k} denote the aligned representation embedding for the kth category hyperspectral pixels, N_{C_k} denote the number of pixels belonging to the kth category, and M_{C_k} denote the stored representation embedded in the memory pool for the kth category. We computed the cosine similarity S_{C_k} between R_{C_k} and M_{C_k}, as depicted in the following equation:

$$\mathbf{S}_{C_k}^{(i)} = \frac{\mathbf{R}_{C_k}^{(i)} \cdot \mathbf{M}_{C_k}}{\|\mathbf{R}_{C_k}^{(i)}\|_2 \cdot \|\mathbf{M}_{C_k}\|_2} \tag{3}$$

where $M_{C_k} \in R^{1\times L}$ and $\|\cdot\|_2$ denotes the L2 norm. We updated the representation of the kth category C_k according to the following equation:

$$\mathbf{R}'_{C_k} = \frac{1}{\sum_{j=1}^{N_{C_k}} \left(1 - \mathbf{S}_{C_k}^{(j)}\right)} \sum_{i=1}^{N_{C_k}} \left(1 - \mathbf{S}_{C_k}^{(i)}\right) \cdot \mathbf{R}_{C_k}^{(i)}. \tag{4}$$

We used a moving average to compute the updated memory for each category, as depicted in the following equation:

$$\mathbf{M}'_{C_k} = (1-m) \cdot \mathbf{M}_{C_k} + m \cdot \mathbf{R}'_{C_k} \tag{5}$$

where m represents the momentum. This updating procedure aids the model to better adapt to the data distribution of the target domain. For clarity, we summarized the category memory pool update process based on the above discussion in Algorithm 1.

Algorithm 1 Algorithm for Updating Category Memory

Input: The representation embeddings R_{C_k} of the patches of the kth category, and the stored representation embedding M_{C_k} of the kth category.

Initialization: The number of training epochs T=100.
for t in range $(0, T)$ **do**
  Compute the cosine similarity S_{C_k} between each category R_{C_k} and the memory R_{C_k} by Eq. (3);
  Update the representation embeddings R_{C_k} by Eq. (4);
  Calculate the updated memory M'_{C_k} of each category by Eq. (5);
  Update the stored representation embeddings M_{C_k};
end for

Output: The updated stored representation embedding M_C.

C. CA Module

The CA module aims to adaptively fuse the category representation embeddings stored in the category representation memory pool into the feature representations of both the source domain and hyperspectral pixels in the target domain. This is done to improve the differentiation between categories. As shown in Fig. 3, the stored category representations and a 1×1 convolution layer are applied to each hyperspectral pixel feature representation to generate three distinct feature maps: $\mathcal{Q} \in R^{1\times B}$, $\mathcal{K} \in R^{K\times B}$, and $\mathcal{V} \in R^{K\times B}$. Subsequently, the $\|$ feature is transposed and matrix-multiplied with the $\mathcal{Q}$ to feature to derive the raw relationship weight matrix. Application of the softmax function to these raw weights normalizes them into a probabilistic distribution form. The normalized weights are then multiplied with the feature $\mathcal{V}$ to obtain the weighted value features. These weighted features

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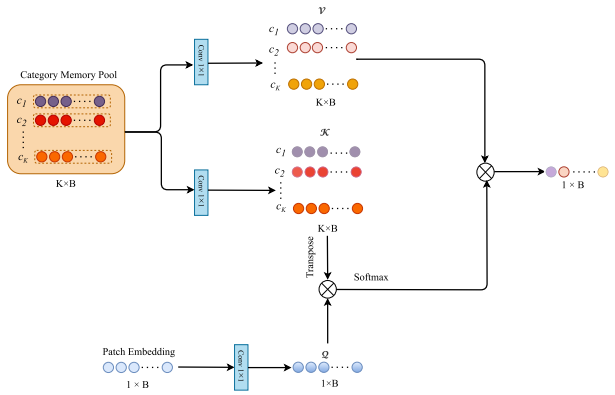


Fig. 3. Structure of CA module.

are concatenated with the original features and then integrated through a 1×1 convolution, batch normalization, and ReLU activation function layer. This fusion helps to mitigate the influence of domain-specific features on representation learning.

D. Adversarial Learning With Biclassifier

During the model transfer process, certain features learned in the source domain might not translate effectively to the target domain, and these features are referred to as hard-to-transfer features. To enhance the discriminative capability of the classifier regarding these hard-to-transfer features and enhance the CSGNet's generalization performance in the target domain, this article utilizes two classifiers, namely, C_1 and C_2, which collectively serve as the discriminator. They undergo training with the encoder E through adversarial learning. The objective is to generate features with high transferability between the source and target domains.

In the UDA problem, ensuring accurate classification of source-domain samples by the classifier is fundamental. Therefore, to fully exploit the available supervised data, we initially trained the entire network on the original domain in each iteration, aiming to reduce the supervised loss of the source-domain data. Specifically, we utilized the cross-entropy loss as the source-domain classification loss and minimized this loss to simultaneously update the encoder E and the two classifiers C_1 and C_2, as depicted in (6) and (7). This process ensures that the model has good feature learning and classification capability in the source domain

$$\min_{\theta_E,\theta_{C_1}} L^s_{C_1}(X_s, T, M, Y_s) = \frac{1}{N_s}\sum_{i=1}^{N_s} L_{\mathrm{CE}}\left(\mathbf{p}^s_{i,1}, \mathbf{y}^s_i\right) \tag{6}$$

$$\min_{\theta_E,\theta_{C_2}} L^s_{C_2}(X_s, T, M, Y_s) = \frac{1}{N_s}\sum_{i=1}^{N_s} L_{\mathrm{CE}}\left(\mathbf{p}^s_{i,2}, \mathbf{y}^s_i\right). \tag{7}$$

Variables θ_E, θ_{C_1}, and θ_{C_2} denote the parameters of the encoder E and the two classifiers C_1 and C_2, respectively. T represents the set of category descriptions, and M is the category memory pool. The expression $p^s_{i,1} = C_1(E(x^s_i, T, M))$ represents the label prediction of the first classifier C_1 for the ith sample in the source domain, while $p^s_{i,2} = C_2(E(x^s_i, T, M))$ represents the label prediction of the second classifier C_2 for the ith sample in the source domain. The term y^s_i denotes the ground-truth label for the ith sample in the source domain, and N_s is the number of training samples in the source domain.

Integrating the above loss functions, the total loss of CSGNet in the source domain is defined as follows:

$$\min_{\theta_{C_1},\theta_{C_2}} L_{\mathrm{source}} = L^s_{C_1} + L^s_{C_2} + \beta L_{\mathrm{align}} \tag{8}$$

where β represents the hyperparameter that balances the image–text alignment loss and α represents the hyperparameter that denotes the proportion of fine-grained text loss within the image–text alignment loss.

To measure the prediction discrepancy between the two classifiers C_1 and C_2 in the target domain, this article cites relevant literature and introduces a classifier prediction discrepancy metric $\mathcal{D}^t_{C_1,C_2}$, as depicted in the following equation:

$$\mathcal{D}^t_{C_1,C_2} = \frac{1}{N_t}\sum_{j=1}^{N_t}\left(\sum_{m,n=1}^{K}\mathbf{A}^{(j)}_{mn} - \sum_{m=1}^{K}\mathbf{A}^{(j)}_{\mathrm{mm}}\right) \tag{9}$$

where $m, n \in \{1, 2, \ldots, K\}$, in which K denotes the number of categories. Matrix A is the prediction correlation matrix between the two classifiers. The matrix $A^{(j)}$denotes the prediction correlation matrix for the jth sample in the target domain. $A^{(j)}{=}p^t_{j,1}p^{t\top}_{j,2}$, $p^t_{j,1}, p^t_{j,2} \in R^{K\times 1}$, where $p^t_{j,1}$ and $p^t_{j,2}$ denote the classification probability predictions by classifiers C_1 and C_2 for the jth sample in the target domain, respectively. The element $A^{(j)}_{mn}$ denotes the product of the probability that classifier C_1 predicts the jth sample in category m and the probability that classifier C_2 predicts the jth sample in category n

$$\min_{\theta_{C_1},\theta_{C_2}} L^s_{C_1}(\mathbf{X}_s, \mathbf{Y}_s) + L^s_{C_2}(\mathbf{X}_s, \mathbf{Y}_s) - \lambda\mathcal{D}^t_{C_1,C_2}. \tag{10}$$

$$\min_{\theta_E} \mathcal{D}^t_{C_1,C_2}. \tag{11}$$

Under source-domain supervision, we first fixed the parameters of the encoder E and optimized the parameters θ_{C_1} and θ_{C_2} of the two classifiers, as shown in (10). The objective is to maximize the prediction discrepancy $D^t_{C_1,C_2}$ between the two classifiers in the target domain while minimizing the fitting error in the source domain. In this case, the two classifiers will attempt to exploit and leverage the domain-specific features that pose challenges for category prediction transfer. Subsequently, we fixed the two classifiers C_1 and C_2 and optimized the parameters θ_E of the encoder E, as shown in (11). During this phase, encoder E leverages an iterative optimization-based adversarial learning strategy to generate cross-domain feature representations with improved transferability. Specifically, the biclassifier architecture (C_1 and C_2) engages in dynamic interaction through prediction discrepancies, thereby ensuring the separation of domain-invariant features from domain-specific ones in the feature space. The goal is to minimize the prediction discrepancy $D^t_{C_1,C_2}$ between the two classifiers in the target domain. To ensure clarity, we have summarized the training process of CSGNet into Algorithm 2 based on the preceding discussions.

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Algorithm 2 Algorithm of CSGNet

Input: Source domain samples $X_s = \{x_i^s, y_i^s\}_{i=1}^{N_s}$; Target domain samples $X_t = \{x_j^t\}_{j=1}^{N_t}$; Category memory pool: M_C; Coarse-grained text: S_{coarse}; Fine-grained text: S_{fine}; max_iteration.

1: Randomly initialize parameters θ_E, θ_{C_1},θ_{C_2}.

2: **repeat**

3: Update θ_E, θ_{C_1}, θ_{C_2} under the source domain X_s supervision by Eq. (6) and Eq. (7).

4: Update θ_E under the supervision of prior knowledge S_{coarse} and S_{fine} of land cover classes by Eq. (2).

5: Update θ_{C_1}, θ_{C_2} by maximizing bi-classifier prediction discrepancy $D^t_{C_1,C_2}$ in the target domain X_t while minimizing the classification loss in the source domain X_s by Eq. (10).

6: Update θ_E by minimizing bi-classifier prediction discrepancy $D^t_{C_1,C_2}$ in the target domain X_t by Eq. (11).

7: **until** max_iteration is reached.

Output: Optimal parameters θ_E^*, $\theta_{C_1}^*$, $\theta_{C_2}^*$ of image encoder E and bi-classifier C_1 and C_2.

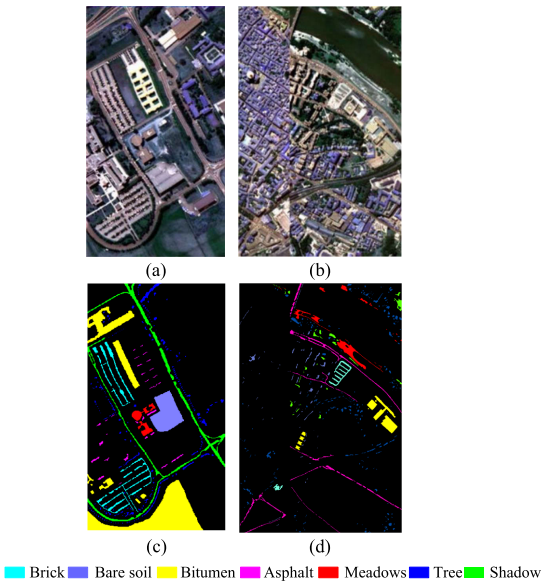


Fig. 4. Pavia dataset. (a) Pavia University pseudo-color image. (b) Pavia Center pseudo-color image. (c) Pavia University ground-truth map. (d) Pavia Center ground-truth map.

IV. Experiments Results and Analysis

In this section, we first introduce the three cross-scene datasets used in this study: the Pavia dataset, the Houston dataset, and the Shanghai–Hangzhou dataset; then, we compare the results of the proposed method with those of six state-of-the-art cross-domain methods on the three cross-scene hyperspectral datasets, i.e., the HTCNN [35], ED-DMM-UDA [40], DSAN [36], CLDA [37], deep cross-domain few-shot learning (DCFSL) [50], language-aware domain generalization network(LDGNet) [27], EHSNet [47], and topological structure and semantic information transfer network (TSTNet) [28]. Finally, we conduct ablation experiments and discuss some of the parameters associated with CSGNet.

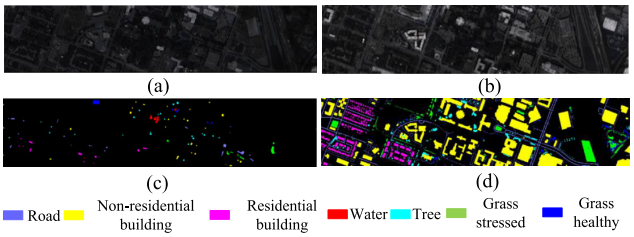


Fig. 5. Houston dataset. (a) Pseudo-color image of Houston 2013. (b) Pseudo-color image of Houston 2018. (c) Ground-truth map of Houston 2013. (d) Ground-truth map of Houston 2018.

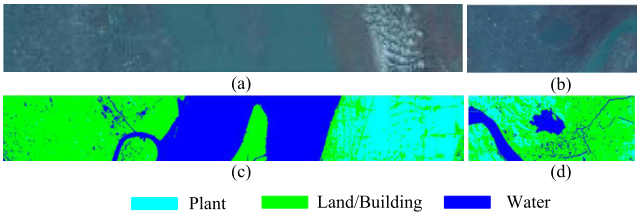


Fig. 6. Shanghai–Hangzhou dataset. (a) Pseudo-color image of Shanghai. (b) Pseudo-color image of Hangzhou. (c) Ground-truth map of Shanghai. (d) Ground-truth map of Hangzhou.

A. Experimental Data

To evaluate the effectiveness of the proposed method, three cross-scene HSI datasets were utilized: the Pavia dataset, the Houston dataset, and the Shanghai–Hangzhou dataset. These datasets were collected by different sensors over various land cover areas. In the following, we provide a detailed introduction to each dataset.

1) Pavia Dataset: The Pavia dataset consists of the University of Pavia (PU, source domain) and Pavia Center (PC, target domain). The ROSIS sensor in the northern Italian city of Pavia collected both. The PC has 1096 × 715 pixels and 102 spectral bands. The PU has 610 × 340 pixels, 103 spectral bands, and 1.3-m spatial resolution, with the last band removed to match the same number of spectral bands as PC. They both share the same seven classes, and the category names and sample counts are listed in Table I. Fig. 4 shows their false-color images and ground-truth maps.

2) Houston Dataset: The Houston dataset includes Houston 2013 (source domain) and Houston 2018 (target domain) images captured on the University of Houston campus in Houston, Texas, USA, using different sensors and years. Houston 2013 contains 349 × 1905 pixels and 144 spectral bands with a spatial resolution of 2.5 m. Houston 2018 contains 209 × 955 pixels and 48 spectral bands with a spatial resolution of 1 m. The spectra have the same wavelength range from 380 to 1050 nm. Specifically, for the Houston 2013 dataset, we selected 209 × 955 overlapping regions and 48 spectral bands corresponding to Houston 2018. Both datasets contain seven classes. The land cover category names and sample counts are shown in Table II. Fig. 5 displays their false-color images and ground-truth maps.

3) Shanghai–Hangzhou Dataset: The Shanghai (source domain) and Hangzhou (target domain) datasets were acquired by the EO-1 Hyperion hyperspectral sensor, which removes corrupted bands and retains 198 spectral bands. Shanghai has

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TABLE I
LAND COVER CLASSES AND SAMPLE COUNTS IN THE PAVIA DATASET

Category		Number of Samples	
No	Name	Source Scene	Target Scene
C1	Tree	3064	7598
C2	Asphalt	6631	9248
C3	Brick	3682	2685
C4	Bitumen	1330	7287
C5	Shadow	947	2863
C6	Meadow	18649	3090
C7	Bare soil	5029	6584
Total		39332	39355

TABLE II
LAND COVER CLASSES AND SAMPLE COUNTS IN THE HOUSTON DATASET

Category		Number of Samples	
No	Name	Source Scene	Target Scene
C1	Grass healthy	345	1353
C2	Grass stressed	365	4888
C3	Trees	365	2766
C4	Water	285	22
C5	Residential building	319	5347
C6	Non-residential buildings	408	32459
C7	Road	443	6365
Total		2530	53200

TABLE III
LAND COVER CLASSES AND SAMPLE COUNTS IN THE SHANGHAI–HANGZHOU DATASET

Category		Number of Samples	
No	Name	Source Scene	Target Scene
C1	Water	123123	18043
C2	Land/Building	161689	77450
C3	Plant	83188	40207
Total		368000	135700

1600 × 230 pixels, while Hangzhou has 590 × 230 pixels. Both datasets share the same three classes: water, ground/building, and vegetation. The category names and sample numbers are presented in Table III. Fig. 6 shows their false-color images and ground-truth maps.

B. Experimental Settings

The proposed CSGNet model is built on the PyTorch deep learning framework. All the experiments were conducted on the Ubuntu 18.04 operating system utilizing a GeForce RTX 2080Ti GPU and an Intel[1] Xeon[1] Silver 4214R CPU for training and testing the network models. Detailed information about the experimental environment is provided in Table VII.

We sampled 80% from the Houston dataset, 5% from the Pavia dataset, and 1% from the Shanghai–Hangzhou dataset, ensuring that the proportions of each category in the sampled data match those in the total samples. The image and text encoders are optimized via adaptive moment estimation (Adam). For weight decay, the default value for L2-norm regularization was set to $1e^{-4}$ for all the modules. For the initialization of the text encoder, we loaded the ViT-B-32.pt model pretrained using CLIP. The HSI with <category name> was used as a template to generate coarse-grained text descriptions for each category. Additionally, fine-grained text descriptions were manually provided, as shown in Tables IV–VI.

[1]Registered trademark.

TABLE IV
FINE-GRAINED TEXTUAL DESCRIPTIONS OF THE PAVIA DATASET

Category	Fine-grained text
Trees	The trees beside road
	The trees appear as small circles
Asphalt	Trees grew along the asphalt road
	The asphalt road appears as an elongated strip shape
Brick	Brick is a kind of road material
	Bricks are generally arranged in strips
Bitumen	Bitumen is a material for building surfaces
	Bitumen is a widely used waterproof material
Shadow	The shadow appears black
	The shadows next to buildings
Meadow	The meadow is pale green
	The surface of the meadow is green with grass
Bare soil	No grass on the surface of bare soil
	The bare soil appears as a grayish-black color

TABLE V
FINE-GRAINED TEXTUAL DESCRIPTIONS OF THE HOUSTON DATASET

Category	Fine-grained text
Grass healthy	The grass healthy is dark
	The grass healthy is next to the road
Grass stressed	The grass stressed is pale green
	The grass stressed is next to the road
Trees	The trees are next to buildings
	The trees appear as small circle
Water	The water has a smooth surface
	The water appears dark blue or black
Residential buildings	Residential buildings are densely packed
	Residential buildings appear as small blocks
Non-residential buildings	The shapes of non-residential buildings areinconsistent
	The surface of the meadow is green with grass
Road	The road appears as an elongated strip shape
	Trees grew along the road

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TABLE VI
FINE-GRAINED TEXTUAL DESCRIPTIONS OF THE SHANGHAI–HANGZHOU DATASET

Category	Fine-grained text
Water	The water has a smooth surface
	The water appears dark blue or black
Land/Building	The Land/Building is along the open coast
	The Land/Building is pale green
Plant	Plant grew along the road
	The Plant appear as elongated strip shape

TABLE VII
EXPERIMENTAL ENVIRONMENT DETAILS

Experimental Environment	Parameters
System	Ubuntu 18.04.5 LTS
CPU	Intel(R) Xeon(R) Silver 4214R CPU @ 2.40GHz
GPU	GeForce RTX 2080Ti
RAM	128GB

For ED-DMM-UDA, the source domain is trained for 1000 epochs to make it sufficiently clustered, and the target domain is trained for 20 epochs. The CLDA method is trained by the SGD optimizer; the weight decay is 0.0005, the learning rate is 0.01, and the confidence operation is repeated a total of four times every 20 epochs to update the network parameters. DCFSL is trained via the Adam optimizer and all convolutional kernels are initialized using the Xavier normalization. The number of training iterations was set to 10 000, and one labeled sample was randomly selected each time to participate in model training with a learning rate of 0.001. For TSTNet, the learning rate is 0.01, the optimizer is SGD, and the weight decay is 0.0001. For LDGNet, the experimental setup is the same as in [27], the learning rate is 0.01, the optimizer is SGD, and the weight decay is 0.0001. For EHSnet, the Adam optimizer is used with a learning rate of 0.01 and a weight decay of 0.0001. Quantitative evaluations of the different methods were performed using metrics such as average accuracy (AA), overall accuracy (OA), and the kappa coefficient. The batch size was set to 36, and the number of epochs was set to 100.

C. Comparison and Analysis

This section presents the classification results of the proposed method on three cross-domain datasets and compares them with those of other existing works. To visually assess the classification results of the above methods, the classification results are plotted in Figs. 7 and 8. The marked pixels are shown as ground truth, and the unlabeled pixels are shown as background.

The first experiment is conducted on the Pavia dataset, and Fig. 7 shows the visual results of all the methods. DAAN typically requires labeled data in both the source and target domains for effective domain adaptation and HTCNN, as a heterogeneous transfer learning method, usually demands a substantial quantity of labeled data. As can be seen from the classification results in Fig. 7(a) and (b), many bitumen samples are misclassified as the asphalt category. The brick category is misclassified as the bare soil category, leading to considerable noise in the classification maps. ED-DMM-UDA, DSAN, CLDA, and DCFSL exhibit suboptimal performance in some categories, such as bitumen, brick, and asphalt. The method CSGNet proposed in this article introduces category labeling knowledge guidance, and semantic knowledge guidance focuses on domain-invariant essential feature information and reduces the influence of interference. This results in a noticeable improvement in the classification of the bitumen category, with a significant reduction in large patches of noise. The outcomes are smoother, and the previously observed misclassifications of bitumen as brick and asphalt have decreased, demonstrating the best overall performance, as shown in Fig. 7(g). Table VIII displays the OA, AA, kappa coefficient, and category-specific classification accuracies achieved by the different methods. Table VIII clearly shows that CSGNet outperforms the other methods in multiple land cover categories, particularly in the "brick" and "bitumen" categories, for which accuracies of 95.42% and 84.66%, respectively, are achieved. In the confusing categories of brick and bitumen, most of the methods do not classify well, and the CSGNet proposed in this article classifies optimally, which also shows that CSGNet has some superiority in distinguishing the challenging land cover categories. The category-label semantic-guided domain adaptation framework CSGNet achieves the highest OA, AA, and kappa values, with values of 88.53%, 87.07%, and 86.19%, respectively. Compared to the LDGNet, the OA is improved by 4.22%, the AA is improved by 1.77%, and the kappa is improved by 4.93%, confirming the effectiveness of the category-label semantic guidance.

The second experiment was conducted on the Houston dataset. This cross-domain dataset poses unique challenges because it involves different sensors, acquisition times, and geographical regions, scattering the source-domain category distribution. Relative to other datasets, Houston is notably challenging to classify. The classification accuracy and visualization results for all methods are shown in Fig. 8 and Table IX. Table IX shows a significant difference in the number of samples between the source and target domains for certain classes. Consequently, clustering methods struggle to perform well. On this dataset, the category-label semantic-guided domain adaptation framework CSGNet achieves an OA of 82.11%, an AA of 76.86%, and a kappa of 69.66%, which are improvements of 3.10%, 4.91%, and 4.68%, respectively, over those of the LDGNet. In addition to using performance metrics to evaluate the experimental results, Fig. 8 displays the classification prediction maps for the target domain. The proposed CSGNet method can correctly classify more samples than the other comparison methods, making a closer match between the predicted result maps and the ground-truth maps.

The third experiment was conducted on the Shanghai–Hangzhou dataset, where the abundant spectral information from numerous bands facilitated the learning of transferable knowledge by the model. CSGNet achieved favorable classification results on this dataset, leveraging

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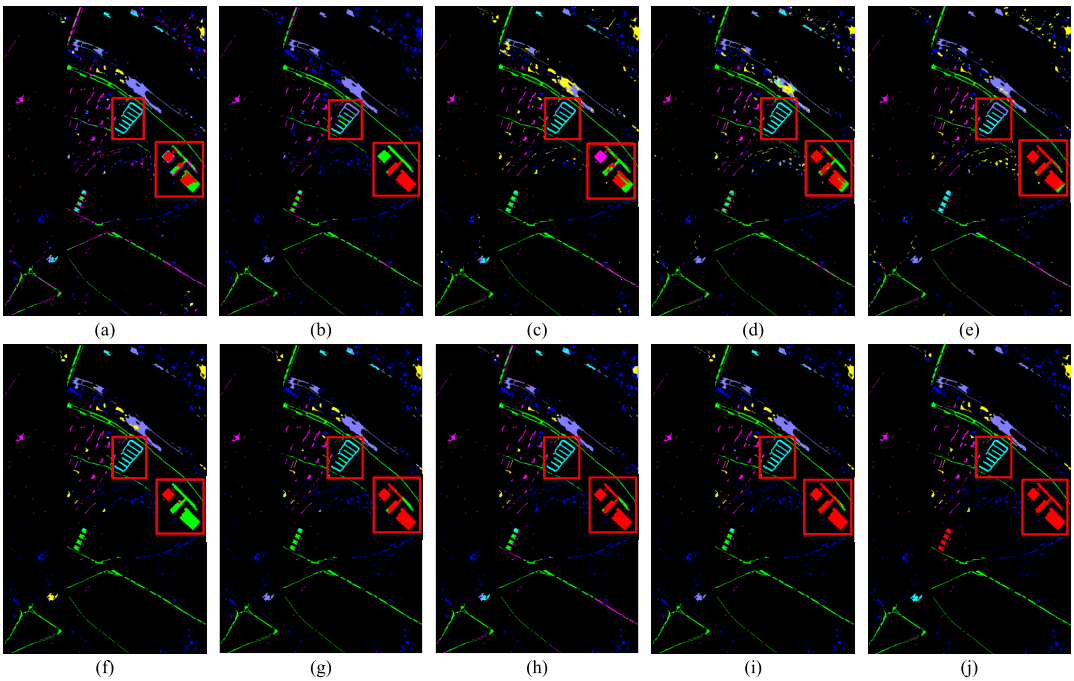


Fig. 7. Visualization of HSI classification results for each model on the target scene Pavia Center obtained with different methods. (a) DCFSL (67.34%). (b) HTCNN (68.00%). (c) TSTNet (71.97%). (d) ED-DMM-UDA (74.35%). (e) DSAN (75.19%). (f) CLDA (78.64%). (g) LDGNet (84.31%). (h) EHSnet (85.04%). (i) CSGNet (88.53%). (j) Ground truth.

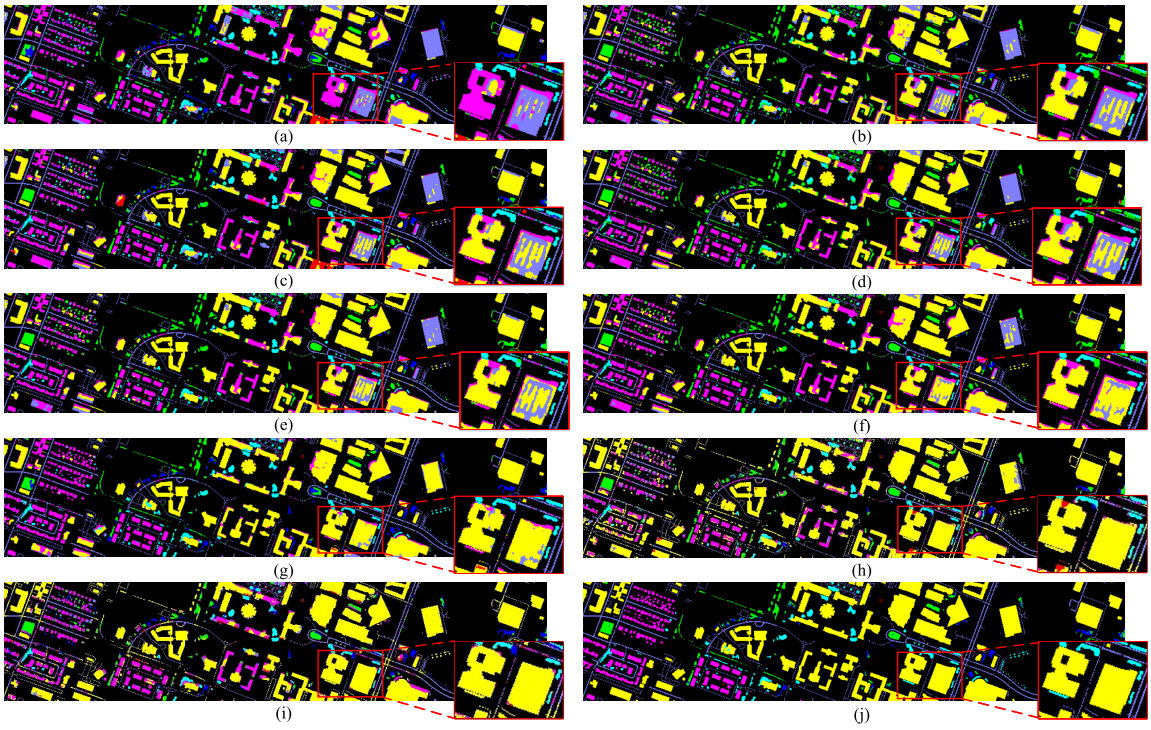


Fig. 8. Visualization of HSI classification results for each model on the target scene Houston 2018 obtained with different methods. (a) DCFSL (57.70%). (b) CLDA (67.68%). (c) ED-DMM-UDA (70.27%). (d) TSTNet (71.29%). (e) DSAN (75.80%). (f) HTCNN (76.71%). (g) LDGNet (79.01%). (h) EHSnet (80.02%). (i) CSGNet (82.11%). (j) Ground truth.

the increased spectral details to extract more discriminative features. The classification accuracy for all methods is shown in Table X. As shown in Table X, CSGNet achieved an OA of 86.43%. Compared to ED-DMM-UDA, DSAN, CLDA, DCFSL, TSTNet, and LDGNet, CSGNet significantly improved classification accuracy, with OA increases of 13.89%, 8.17%, 11.29%, 13.38%, 4.33%, and 2.59%, respectively.

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TABLE VIII
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE TARGET SCENE PAVIA CENTER

Category	DCFSL [50]	HTCNN [35]	TSTNet [28]	ED-DMM-UDA [40]	DSAN [36]	CLDA [37]	LDGNet [27]	EHSnet [47]	CSGNet
Tree	67.89	84.32	77.73	67.91	58.80	98.66	93.80	86.34	93.70
Asphalt	62.22	91.24	93.23	95.24	80.54	98.89	79.44	85.36	93.35
Brick	90.76	43.61	82.94	77.65	45.66	78.51	90.20	66.16	95.42
Bitumen	43.05	30.16	33.64	64.75	77.48	17.22	80.79	89.52	84.66
Shadow	98.29	84.28	99.51	100	99.48	91.48	96.93	91.01	82.33
Meadow	53.53	85.86	96.99	51.55	61.33	77.99	75.76	71.47	72.49
Bare soil	84.25	76.66	58.96	61.28	92.04	89.82	80.21	89.11	87.52
OA (%)	67.34 ± 1.82	68.00 ± 2.18	71.97 ± 1.52	74.35 ± 1.46	75.19 ± 1.34	78.64 ± 1.65	84.31 ± 1.36	85.04 ± 1.49	**88.53±1.25**
AA (%)	71.43 ± 1.71	64.69 ± 1.05	70.88 ± 1.66	74.06 ± 1.64	73.62 ± 1.65	78.94 ± 1.36	85.30 ± 1.61	81.38 ± 1.57	**87.07±1.39**
Kappa×100	61.51 ± 1.25	63.68 ± 1.69	67.96 ± 2.04	69.26 ± 2.45	70.39 ± 1.57	73.96 ± 1.87	81.26 ± 1.44	81.90 ± 1.68	**86.19±1.30**

TABLE IX
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE TARGET SCENE HOUSTON 2018

Category	DCFSL [50]	CLDA [37]	ED-DMM-UDA [40]	TSTNet [28]	DSAN [36]	HTCNN [35]	LDGNet [27]	EHSnet [47]	CSGNet
Grass healthy	77.14	71.03	49.30	86.25	60.35	52.83	56.91	65.71	67.04
Grass stressed	44.87	87.97	90.37	77.27	75.20	70.25	84.47	70.76	80.87
Trees	49.87	72.67	72.52	59.22	79.25	54.99	65.55	62.70	65.69
Water	23.81	36.36	90.91	86.36	98.76	56.79	95.45	74.26	100
Residential Buildings	68.43	95.70	81.54	81.50	85.34	55.60	67.89	63.71	81.75
Nonresidential Buildings	61.25	58.07	63.36	73.63	79.65	89.76	88.85	88.06	90.19
Road	39.84	74.82	84.04	48.19	50.37	47.47	44.49	70.47	52.51
OA (%)	57.70 ± 2.25	67.68 ± 1.82	70.27 ± 1.45	71.29 ± 1.29	75.80 ± 1.97	76.71 ± 1.95	79.01 ± 1.53	80.02 ± 1.46	**82.11±1.46**
AA (%)	52.17 ± 1.95	70.95 ± 1.36	76.01 ± 1.43	73.20 ± 1.67	75.56 ± 1.83	61.10 ± 2.15	71.95 ± 1.72	70.81 ± 1.49	**76.86±1.25**
Kappa×100	43.88 ± 1.75	57.11 ± 1.92	57.59 ± 2.27	59.36 ± 1.85	64.10 ± 2.95	63.94 ± 2.62	64.98 ± 2.12	69.15 ± 1.79	**69.66±1.36**

TABLE X
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE TARGET SCENE HANGZHOU

Category	ED-DMM-UDA [40]	DCFSL [50]	CLDA [37]	HTCNN [35]	DSAN [36]	TSTNet [28]	LDGNet [27]	EHSnet [47]	CSGNet
Water	86.20	98.15	99.93	99.67	98.64	86.80	88.50	87.31	89.00
Land/Building	76.52	71.01	81.20	78.34	80.10	86.47	89.63	87.09	91.89
Plant	58.74	65.73	52.35	62.22	65.59	71.60	70.61	76.71	74.75
OA (%)	72.54 ± 1.76	73.05 ± 1.83	75.14 ± 2.18	76.40 ± 1.95	78.26 ± 1.89	82.10 ± 1.52	83.84 ± 1.42	84.04 ± 1.38	**86.43±1.25**
AA (%)	73.82 ± 2.32	78.30 ± 2.15	77.83 ± 1.93	80.08 ± 1.69	81.44 ± 1.76	81.62 ± 1.23	76.71 ± 1.53	83.70 ± 1.46	**85.21±1.24**
Kappa×100	59.85 ± 2.56	58.15 ± 2.23	60.03 ± 1.85	61.91 ± 1.68	64.27 ± 1.28	69.17 ± 1.45	71.02 ± 1.35	74.81 ± 1.51	**75.62±1.32**

D. Analysis of Visual-Linguistic Feature Alignment

To verify the effectiveness of the visual-linguistic feature alignment during training, a visual comparison of the category separability before and after visual-linguistic alignment is presented, as shown in Fig. 9. In this analysis, we use t-distributed stochastic neighbor embedding (t-SNE) for feature projection to visualize the feature distributions from three datasets: Houston, Pavia, and Shanghai–Hangzhou, before and after the visual-linguistic alignment process. For example, in the Houston dataset, before visual-linguistic alignment, the features from the "Grass healthy" and "Trees" categories are mixed, making it difficult to distinguish them. After alignment, the features become clearly separated, with "Grass healthy" positioned distinctly from "Trees," reducing misclassifications. Additionally, after the alignment, the visual features are projected into the textual semantic space, and we can observe that the visual feature embeddings are no longer scattered but now cluster around the corresponding text embeddings, leading to better-defined category separation. In the Pavia dataset, before the alignment, the "Asphalt" and "Bitumen" categories overlap significantly, causing confusion. After the alignment, the feature space becomes more distinct, with "Asphalt" and "Bitumen" positioned separately. Similar to the Houston dataset, the visual features become more tightly aligned with their respective text embeddings, improving the overall clarity of category separability. In the Shanghai–Hangzhou dataset, before alignment, the "Water" and "Land/Building" categories have overlapping features, especially in areas where water bodies and land surfaces are adjacent. After applying visual-linguistic alignment, the separation between "Water" and "Land/Building" is much

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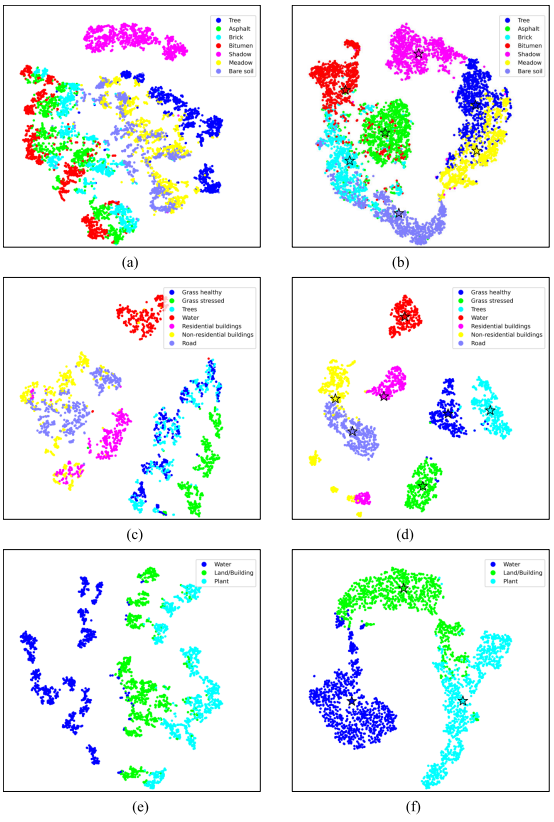


Fig. 9. Visual analysis of category separability before and after visual-linguistic alignment on three datasets. The first column is the distribution of features before visual-linguistic alignment and the second column is the distribution of features after visual-linguistic alignment. Different colors represent different categories. (a) Pavia dataset before VL alignment. (b) Pavia dataset after VL alignment. (c) Houston dataset before VL alignment. (d) Houston dataset after VL alignment. (e) S–H dataset before VL alignment. (f) S–H dataset after VL alignment.

clearer. The visual features now align more closely with their respective category descriptions, resulting in better differentiation. This improvement indicates that the alignment not only reduces confusion but also enhances the grouping ability of the model.

E. Effectiveness Analysis of CA

Features are projected using the t-SNE clustering technique to verify the effectiveness of the CA during training. In the three datasets, the features before and after the CA are visualized, and the comparison results are shown in Fig. 10. Different colors represent different classes, and it can be observed that in Fig. 10(a), (c), and (e), the distribution between classes is chaotic, and there may be considerable overlapping features between different classes with many poorly demarcated regions. After the CA, the domain-invariant features stored in the category memory guide the current image input image features, and a more consistent feature representation is established between the source and target domains, as shown in Fig. 10(b), (d), and (f). The aggregation of features belonging to the same category is greatly improved, the distance between the feature vectors corresponding to the different classes is significantly increased, and the separability is greatly improved.

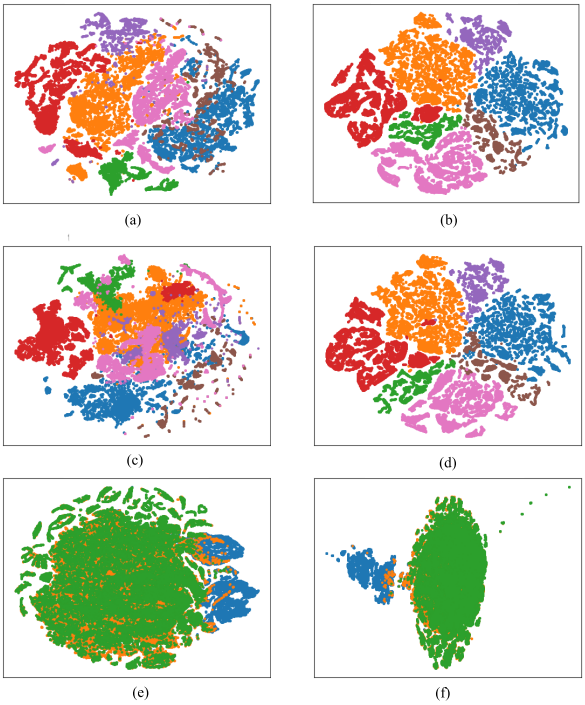


Fig. 10. Visual analysis of category separability before and after CA on three datasets. The first column is the distribution of features before CA and the second column is the distribution of features after CA. Different colors represent different categories. (a) Pavia dataset before CA processing. (b) Pavia dataset after CA processing. (c) Houston dataset before CA processing. (d) Houston dataset after CA processing. (e) S–H dataset before CA processing. (f) S–H dataset after CA processing.

F. Analysis of Hyperparameters

CSGNet contains three hyperparameters that require manual tuning. A parameter sensitivity analysis is conducted to evaluate the sensitivity of CSGNet, as shown in Tables XI–XIII. The fine-grained linguistic feature contribution α, visual-linguistic features alignment loss parameter β, and biclassifier prediction discrepancy regularization parameter λ regarded as adjustable hyperparameters are selected from $\{0.1, 0.3, 0.5, 0.7, 0.9\}$, $1e^{-3}$, $1e^{-2}$, $1e^{-1}$, $1e^{+0}$, $1e^{+1}$, and $5e^{-4}$, $1e^{-3}$, $5e^{-3}$, $1e^{-2}$, $5e^{-2}$, respectively.

Table XI provides the classification results corresponding to different fine-grained linguistic feature contribution α in three datasets. With respect to three datasets, the optimal α is 0.5 for the Houston dataset and the Shanghai–Hangzhou dataset and 0.3 for the Pavia dataset. Tables XII and XIII include the OA of all the experimental datasets for visual-linguistic features alignment loss parameter β and biclassifier prediction discrepancy regularization parameter λ, respectively. The optimal β is $1e^{-1}$ for the Houston dataset and Pavia dataset and $1e^{+0}$ for the Shanghai–Hangzhou dataset; and λ is $1e^{-3}$ for the Houston dataset, $5e^{-2}$ for the Pavia dataset, and $5e^{-3}$ for the Shanghai–Hangzhou dataset.

G. Analysis of Fine-Grained Text Sensitivity

The fine-grained text is mainly used to describe the a priori knowledge of the appearance, color, shape, size, texture, and distribution attributes of ground objects in remote sensing

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TABLE XI
OVERALL CLASSIFICATION ACCURACY (%) WITH DIFFERENT WEIGHTS α FOR THE PROPOSED CSGNET USING THE THREE EXPERIMENTAL DATA

Dataset	Weights α				
	0.1	0.3	0.5	0.7	0.9
Houston	80.98	82.01	**82.11**	80.88	79.95
Pavia	87.42	**88.53**	88.12	87.21	86.16
Shanghai-Hangzhou	80.60	84.52	**86.43**	83.26	85.43

TABLE XII
OVERALL CLASSIFICATION ACCURACY (%) WITH DIFFERENT WEIGHTS β FOR THE PROPOSED CSGNET USING THE THREE EXPERIMENTAL DATA

Dataset	Weights β				
	$1e{-}3$	$1e{-}2$	$1e{-}1$	$1e{+}0$	$1e{+}1$
Houston	81.45	80.81	**82.11**	81.55	81.20
Pavia	85.21	87.03	**88.53**	85.03	86.57
Shanghai-Hangzhou	86.22	85.65	86.14	**86.43**	86.08

TABLE XIII
OVERALL CLASSIFICATION ACCURACY (%) WITH DIFFERENT REGULARIZATION PARAMETERS λ FOR THE PROPOSED CSGNET USING THE THREE EXPERIMENTAL DATA

Dataset	Regularization parameters λ				
	$5e{-}3$	$1e{-}3$	$5e{-}3$	$1e{-}2$	$5e{-}2$
Houston	81.36	**82.11**	80.66	69.46	58.46
Pavia	83.96	85.03	86.64	87.31	**88.53**
Shanghai-Hangzhou	85.72	86.16	**86.43**	83.38	83.41

TABLE XIV
ABLATION STUDY RESULTS OF CSGNet ON THREE HSI DATASETS

VL	BCAL	CA	Houston	Pavia	Shanghai-Hangzhou
			75.83	81.27	81.63
✓			78.80	84.15	83.54
	✓		78.18	84.69	84.23
		✓	77.03	82.31	83.48
✓	✓		81.37	86.97	85.97
	✓	✓	79.21	86.08	84.88
✓		✓	79.84	85.17	84.71
✓	✓	✓	**82.11**	**88.53**	**86.43**

images, as well as the spatial topological relationships between them in the proposed CSGNet. If the semantic descriptions of the categories are incorrect, it may lead to a bias in the category representations learned by the model, which in turn affects the domain adaptation capability of the proposed model. Table XVI contains artificially defined fine-grained text, containing strong related descriptions, weakly related descriptions, and irrelevant descriptions.

1) *Strong Related Descriptions:* The name of each land cover class is entered as a keyword, and sentences related to the attributes of the land cover class are selected from Google.
2) *Weakly Related Descriptions:* Sentences are randomly generated for each land cover class, and the sentences are not related to the attributes of the land cover class.
3) *Irrelevant Descriptions:* Illogical sentences are randomly generated for each land cover class.

The following experiments are conducted using the Houston dataset as an example to analyze the sensitivity of CSGNet to fine-grained text. The classification accuracy is shown in Table XVII. Compared to using strong related descriptions, when weakly related descriptions are used as fine-grained text, overall classification accuracy is decreased by 0.99%, 0.19%, and 0.82%. Also, when irrelevant descriptions are used as fine-grained text, overall classification accuracy is decreased by 1.64%, 7.60%, and 3.98%. This shows that related fine-grained textual descriptions perform better than irrelevant textual descriptions.

This article tested the effect of different text settings on model performance and found that the semantic information described in the fine-grained text that is not related to the category has a large impact on model performance, but the semantic information described that is weakly related to the category does not have a significant impact on model performance. The experimental results show that CSGNet is robust to different text settings.

H. Analysis of Time Costs

Table XV compares training times, inferring times, inference parameters, and OA between our proposed method and the comparison methods on three datasets. During the training process, due to the use of coarse-grained text and fine-grained text, CSGNet performs more complex calculations and uses more data during model training and therefore takes longer than other methods in training time. In terms of inference parameters, CSGNet uses significantly fewer parameters than the other methods, suggesting that CSGNet is more lightweight and conducive to deployment in resource-limited environments. Meanwhile, CSGNet achieves the highest OA on all three datasets, outperforming LDGNet by 4.19%, 3.10%, and 2.59%. In summary, CSGNet performs better in terms of OA. The training and inference time does not lag behind that of other comparative models. This indicates that CSGNet strikes a good balance between efficiency and performance.

I. Ablation Experiments

To further demonstrate the effectiveness of the visual-linguistic features alignment (VL), BCAL, and CA module in CSGNet, ablation experiments were conducted by removing specific components for comparison. The performances of the models on the Pavia, Houston, and Shanghai–Hangzhou datasets were analyzed and compared, and the quantitative results of the ablation experiments are presented in Table XIV.

The results show that the complete CSGNet model performs the best on all the datasets. When CSGNet removes the VL, the OA decreases by 2.90%, 2.45%, and 1.55% on the three datasets. This demonstrates that the prior knowledge of coarse-grained text descriptions and fine-grained text descriptions for each category can help CSGNet enhance feature representation capabilities, enhancing the semantic expression capability of feature learning. When CSGNet removes the BCAL, the OA decreases by 2.27%, 3.36%, and 1.72%, on the three datasets. This demonstrates that BCAL can enhance the

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TABLE XV
COMPARISON OF THE PERFORMANCE OF DIFFERENT METHODS IN TERMS OF TRAINING TIME, INFERENCE TIME, AND PARAMETERS

Datasets	Metrics	HTCNN [35]	ED-DMM-UDA [40]	DSAN [36]	CLDA [37]	DCFSL [50]	TSTNet [28]	LDGNet [27]	EHSnet [47]	CSGNet
Pavia	Training time (s)	280.11	321.73	226.30	289.54	628.62	534.22	432.60	485.99	957.30
	Inference time (s)	4.23	5.19	3.18	1.55	15.23	19.57	15.23	29.39	12.51
	Params (M)	2.83	0.18	25.18	0.18	1.87	7.43	1.67	1.59	0.91
	OA (%)	68.00	74.35	75.19	78.64	67.34	71.97	84.34	85.04	88.53
Houston	Training time (s)	292.31	340.05	240.12	301.23	652.13	474.86	417.44	397.34	767.16
	Inference time (s)	4.42	5.95	3.88	2.06	8.16	15.47	17.27	33.27	17.16
	Params (M)	2.82	0.17	25.02	0.18	1.83	7.42	0.78	0.86	0.47
	OA (%)	76.71	70.27	75.80	67.68	57.70	71.29	79.01	80.02	82.11
Shanghai-Hangzhou	Training time (s)	206.31	450.75	430.81	198.05	583.22	843.22	635.72	859.51	906.31
	Inference time (s)	13.68	15.33	9.25	7.01	18.68	116.26	67.97	80.97	52.11
	Params (M)	2.85	0.20	28.37	0.19	2.25	7.45	3.17	3.59	1.66
	OA (%)	76.40	72.54	78.26	75.14	73.05	82.10	83.84	84.04	86.43

TABLE XVI
FINE-GRAINED TEXTUAL DESCRIPTIONS WITH DIFFERENT RELEVANCE OF THE HOUSTON DATASET

Category Name	Fine-grained Text Description
Grass healthy (Strong related)	The grass healthy is lush and vibrant.
	Healthy grass has a rich green hue.
Grass healthy (Weak related)	This is an image of healthy grass.
	This piece of grassland looks very healthy.
Grass healthy (Irrelevant)	The silent observer reflects upon the bustling avenue.
	The heartbreaking profession excites into the abaft guarantee.

TABLE XVII
OA OF FINE-GRAINED TEXT DATASET

Fine-grained Text	Overall Accuracy (OA)		
	Houston	Pavia	Shanghai-Hangzhou
Strong related	**82.11**	**88.53**	**86.43**
Weakly related	81.12	88.34	85.61
Irrelevant	80.47	80.93	82.45

discriminative capability of the classifier regarding these hard-to-transfer features and enhance the model's generalization performance. When the CA is removed from CSGNet, the OA decreases by 0.74%, 1.56%, and 0.46% on the three datasets. This underscores the crucial role of the CA in domain adaptation, as it stores the inherent invariant feature representations of categories, mitigating the impact of domain distribution differences. Furthermore, when all three modules are removed, the OA decreased by 6.28%, 7.26%, and 4.80% on the three datasets, indicating a significant decrease in performance when all modules are removed and emphasizing the importance of their synergistic effect.

V. CONCLUSION

In this article, we propose CSGNet, a new domain adaptation network for cross-domain HSI classification, which establishes a more consistent category feature representation between the source and target domains. To incorporate category prior knowledge during feature learning, we extracted semantic features from the textual descriptions associated with each category. We then aligned the visual features with the linguistic features in the semantic space and stored the visual-linguistic features in the category representation memory pool. To eliminate the domain hard-to-transfer features and retain the cross-domain invariant features, we propose a BCAL method to enhance the classifier's ability to distinguish those hard-to-transfer features, thereby improving the encoder's capacity to generate cross-domain invariant features in the unlabeled target domain. The proposed CA module improves CSGNet's performance by integrating representations from various categories. This focuses the model on acquiring cross-domain invariant representations for each hyperspectral pixel, ensuring robust and effective learning across domains. Finally, extensive experimental results on three cross-domain datasets validate the effectiveness of the proposed CSGNet in HSI domain adaptation.

In future work, we aim to expand CSGNet to cross-domain HSI classification on open sets by incorporating the idea of unknown category processing. Additionally, we will explore advanced methods for incorporating more prior knowledge into the CSGNet, which can help identify new land cover types and enhance the effectiveness of the task.

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